

**NAME**

saw – get the standard atomic weight

**LIBRARY**

ciaaw library (*libciaaw*, *-lciaaw*)

**SYNOPSIS**

```

Fortran      function saw(s,ab,u)result(res)
               character(len=*), intent(in) :: s
               logical, intent(in), optional :: ab
               logical, intent(in), optional :: u
               real(dp) :: res

C            double ciaaw_saw(char *s,int n,bool ab,bool u);

Python      saw(s:str,abridged:bool=True,uncertainty:bool=False)->float:

```

**DESCRIPTION**

Get the standard atomic weight for the element *s*. The default behavior is to provide the abridged value, *ab* is *TRUE*, recommended by the CIAAW. The uncertainty can be retrieved by setting *u*=*TRUE*.

Arguments:

|           |                                                                             |
|-----------|-----------------------------------------------------------------------------|
| <i>s</i>  | Element symbol.                                                             |
| <i>ab</i> | Set to False if the abridged value is not desired. Default to <i>TRUE</i> . |
| <i>u</i>  | Set to True if the uncertainty is desired. Default to <i>FALSE</i> .        |

**RETURN VALUE**

|     |                                       |
|-----|---------------------------------------|
| NaN | If the provided element is incorrect. |
| -1  | If the element does not have a SAW.   |

**EXAMPLES**

```

Fortran:
    use ciaaw
    saw("H", ab=.false., u=.true.)

C:
    include <stdio.h>
    include "ciaaw.h"
    ciaaw_saw("H", 1, false, true)

Python:
    import pyciaaw

```

```
pyciaaw.saw("H", ab=False, u=True)
```

**SEE ALSO**

***ciaaw*(3), *ciaaw\_cli*(1), *ciaaw\_version*(3), *ciaaw\_saw*(3), *ciaaw\_ice*(3), *ciaaw\_nice*(3), *ciaaw\_naw*(3), *ciaaw\_nnaw*(3), *ciaaw\_ices*(3), *ciaaw\_naws*(3), *ciaaw\_s2z*(3)**